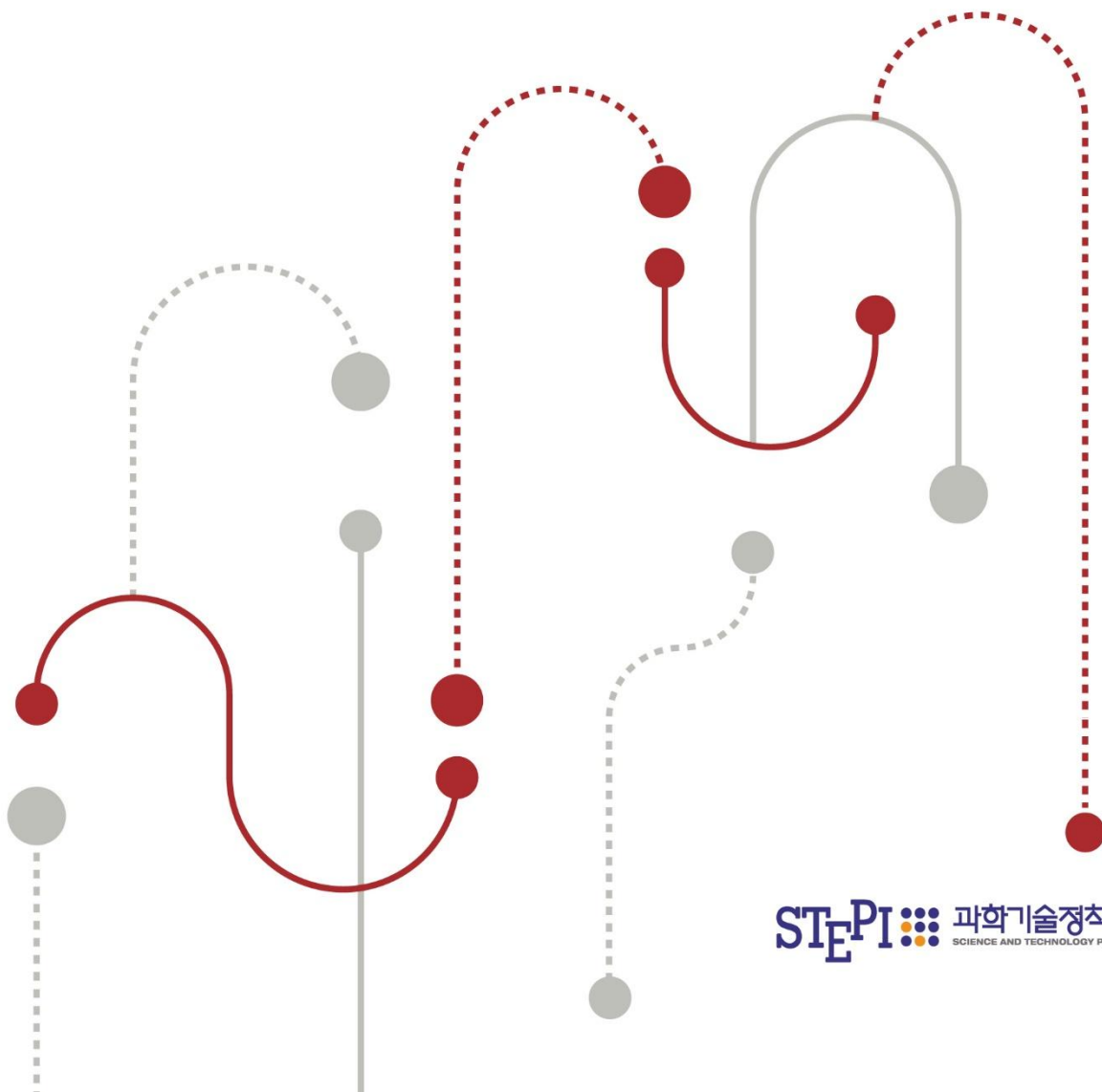


SCIENCE & TECHNOLOGY POLICY

STEPI Insight

Determinants of Earnings among Recent STEM PhD Graduates in Korea and Policy Implications



| Vol. 355. February 19th. 2026 |

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SUMMARY

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1. Research Background and Objectives

- **Recent science and engineering PhD graduates are core resources for national R&D competitiveness; understanding their compensation structure and early-career earnings is essential**
 - Science and engineering PhD holders play central roles in innovation by conducting R&D and training the next generation of researchers; their effective training and utilization underpin national R&D performance
 - The doctoral program is a high-cost human capital investment requiring long-term intensive investment from both individuals and the nation, necessitating the establishment of efficient investment directions to ensure appropriate compensation in the labor market
- **This study provides empirical evidence relevant for doctoral training and STEM workforce policy by analyzing the correlates of earnings among recent science and engineering PhD graduates**
 - We examine associations between earnings and field of study, doctoral program characteristics, research outputs, and research experiences, and document the magnitude of earnings differentials across groups

- We compare gross monthly earnings and hours-adjusted hourly wages to diagnose whether observed earnings gaps primarily reflect working-hour differences
- We further assess whether institutional mobility between the bachelor's and PhD-granting institutions is associated with additional earnings beyond PhD institution type
- The results identify human-capital attributes that receive high market value and provide an empirical basis for individual career strategies and policy design

2. Data Characteristics and Sample Description

■ **To approximate the net association between doctoral training and earnings, the analytic sample is restricted to individuals who pursued the PhD full-time and are currently regular wage-and-salary employees (N=296)**

- The Science and Engineering PhD Follow-up Survey uses a sophisticated sampling design to track degree acquisition and early career paths of science and engineering PhD recipients who graduated from general graduate schools in Korea
- Restricting the sample reduces confounding from prior full-time employment and focuses on human capital accumulated during doctoral training, though external validity is limited to this subgroup

■ **Earnings determinants are organized into five categories: control variables, field of study, doctoral program, research output, and research experience**

- Control variables (e.g., gender, age, and employer sector) capture structural labor-market differences while holding constant human-capital correlates
- Field of study includes academic discipline and whether the field falls under the 12 National Strategic Technologies; doctoral program variables include PhD institution type, integrated master's-doctoral status, time to degree, and degree acquisition costs

- Research output covers academic outputs (publication records) and technological outputs (patent applications, technology transfer); research experience includes participation in R&D projects, workforce development programs, industry-academia collaboration, and international collaboration
- **The sample is predominantly male (63.9%), private sector (72.3%), and engineering majors (72.6%), with earnings gaps across characteristics and academic outputs centered on SCIE-indexed publications**
- Mean age is 33.1 years and mean time to degree is 5.8 years, indicating relatively homogeneous demographic and training characteristics
 - Average earnings differ by gender, employer sector, field of study, doctoral program, and research experience; academic output shows a high SCIE-indexed publication share (82.7%), while technological achievements are mostly absent
 - While 96.9% have R&D project participation experience, only 7.8% report international collaboration experience

3. Empirical Analysis Results

- **PhD-granting institution type is the most critical correlate of earnings, and qualitative indicators of field of study and research output are also significantly associated with earnings**
- Hierarchical OLS results show that PhD institution type increases adjusted R^2 by approximately 9.4 percentage points, the largest incremental contribution among all covariate blocks
 - Academic discipline and National Strategic Technology status increase adjusted R^2 by 6.2 and 3.5 percentage points, respectively, indicating substantial explanatory power of field of study

- Among research outputs, only the share of first-authored publications in SCIE-indexed journals increases adjusted R^2 by approximately 1.6 percentage points

■ **Large earnings differentials exist by institution type and field of study, while gender and employer sector gaps in monthly earnings are largely attributable to working hours**

- Relative to elite research-intensive universities, graduates from small-to- medium Seoul metropolitan private universities earn approximately 14.8% lower gross monthly earnings, large Seoul metropolitan private universities earn 11.8% lower, and regional flagship national universities earn 28.2% lower
- Engineering, National Strategic Technology fields, and integrated master's- doctoral completers show earnings premiums of 12.2%, 13.0%, and 8.9%, respectively; a 1 percentage-point increase in the share of first-authored SCIE-indexed publications is associated with approximately 0.2% higher monthly earnings
- Women earn approximately 11.2% lower monthly earnings than men, and the public sector earns approximately 15.1% lower than the private sector; however, the gender gap becomes statistically insignificant for hourly wages, and the sector gap shrinks substantially to approximately 1.0%

■ **Institutional mobility between bachelor's and PhD-granting institutions is not associated with additional earnings, and PhD institution type has higher explanatory power for earnings than bachelor's institution type**

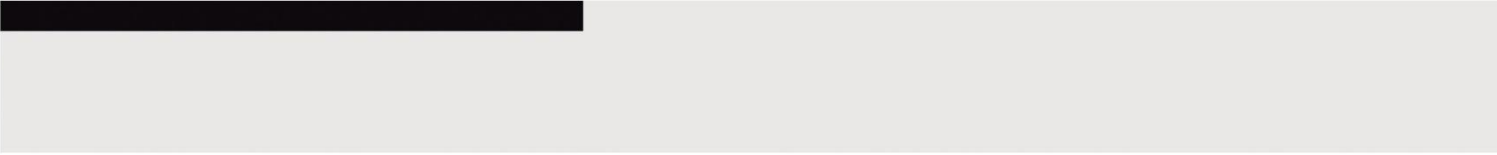
- Mobility follows a consistent pattern toward higher-earning institution types, and mobility rates are higher among lower-earning institution types
- Models based on bachelor's institution type show approximately 20.2% lower explanatory power than PhD-based models, indicating that the PhD institution type is more informative for earnings

- Changes in institution type between the bachelor's and PhD degrees are not statistically associated with earnings beyond the level effect of the PhD institution type

4. Conclusions and Implications

■ This study provides empirical evidence for doctoral training and STEM workforce policy by analyzing earnings correlates of recent science and engineering PhD graduates from multiple perspectives

- PhD institution type is the most critical correlate of earnings, and there is no additional earnings association from institutional mobility between the bachelor's and PhD degrees
 - Further research is needed to investigate the underlying causes of earnings gaps across institution types, and competitiveness disparities between institution types should be mitigated through specialization aligned with each type's unique role
 - Because bachelor's institution type does not appear to constrain outcomes, doctoral enrollment can support upward mobility in labor-market outcomes for students from less-advantaged institutions
- Earnings gaps exist across disciplines and National Strategic Technology fields show earnings premiums; among research outputs, only the share of first-authored SCIE-indexed publications is significantly associated with earnings
 - The earnings premium in National Strategic Technology fields supports the effectiveness of workforce policies in these areas, while the structural earnings disadvantage of natural sciences suggests the need for policy attention to basic science
 - Quantitative research output, such as the number of publications, does not significantly affect earnings; only first-authored research output is valued in the labor market

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- Gender and employer-type earnings gaps are largely influenced by working hours, highlighting the need to diagnose structural gaps through the lens of working-hour differences
 - Because the monthly gender earnings gap becomes insignificant in hourly wages, further analysis is needed to assess whether working-hour differences reflect voluntary choices or structural constraints (e.g., childcare and caregiving burdens)
 - Employer-type earnings gaps vary by earnings measure; the private sector's monthly earnings advantage is largely attributable to longer working hours, and universities show a substantial disadvantage in hourly wages
 - Limitations and future research: the analysis is based on cross-sectional data and documents associations; future work leveraging the panel structure of the survey is needed to distinguish training effects from selection and to strengthen causal interpretation
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